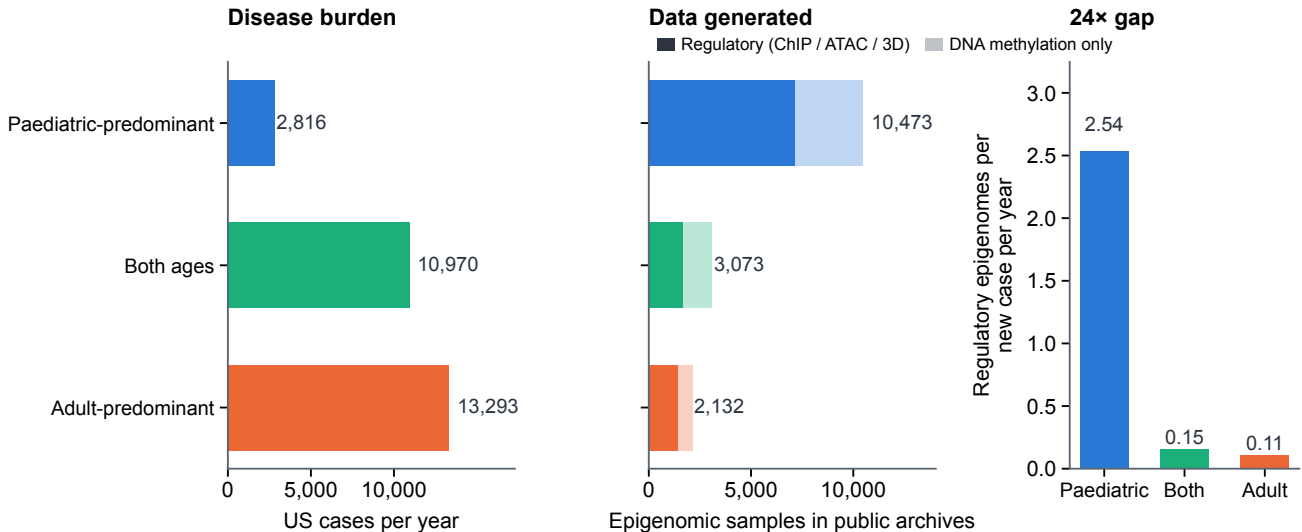
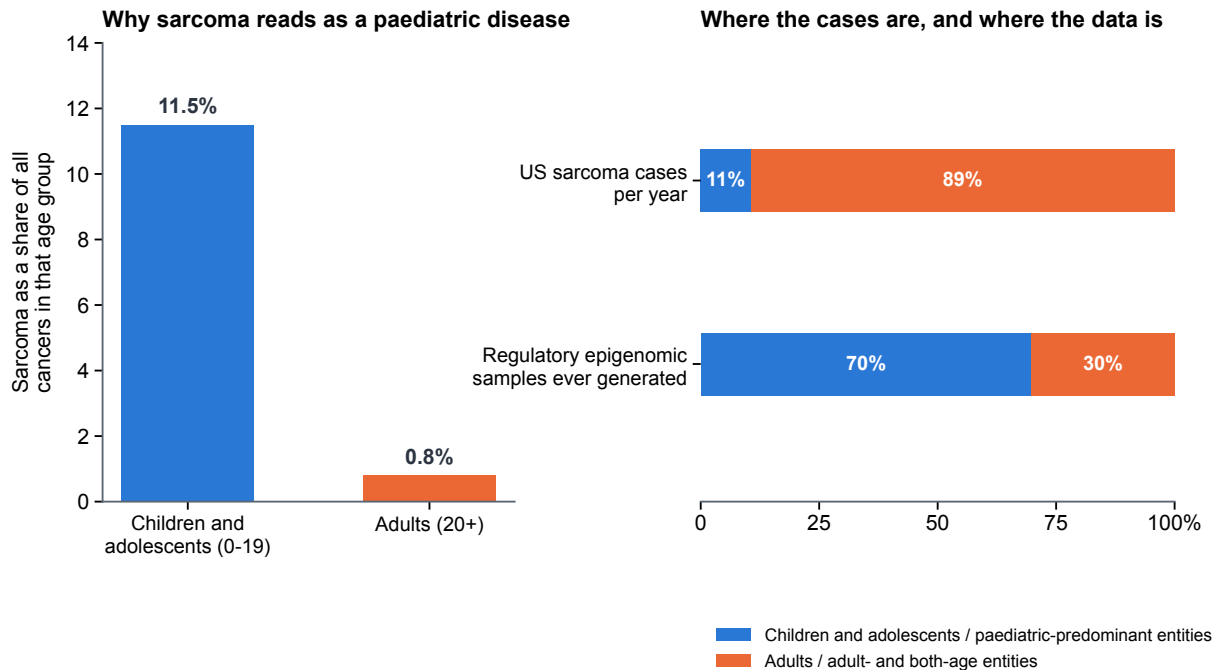


Adult sarcomas carry ~5× the disease burden and a small fraction of the epigenomic data



Sarcoma is 1% of adult cancer and 11% of childhood cancer — and the epigenomic literature followed the 11%



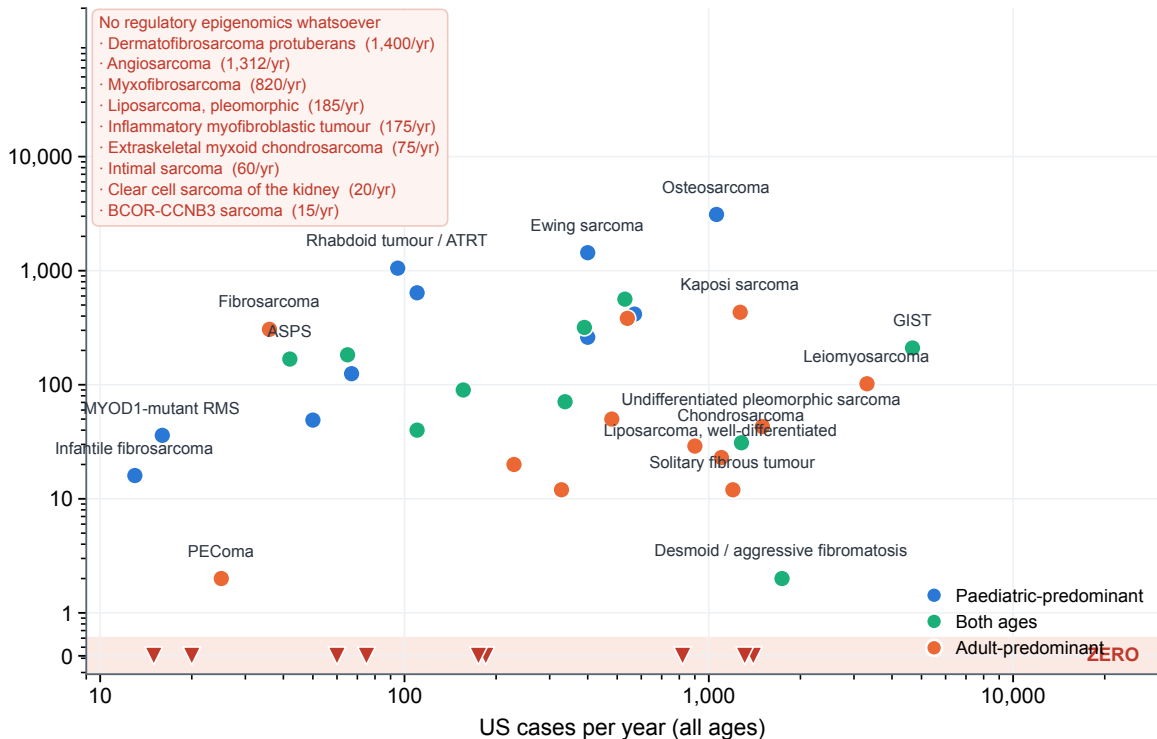
Sources: sarcoma share of all cancers and US case counts, Siegel et al., Cancer Statistics 2026 (PMID 41528114; 4,110 bone + 13,910 soft tissue = 18,020) and SEER 21 age distribution; paediatric share of childhood malignancy, Siegel DA et al., JNCI 2023 (PMID 37433078; ICCG groups VIII + IX = 11.5%). Sample counts from this atlas (T13).

Regulatory epigenomics tracks research history, not disease burden

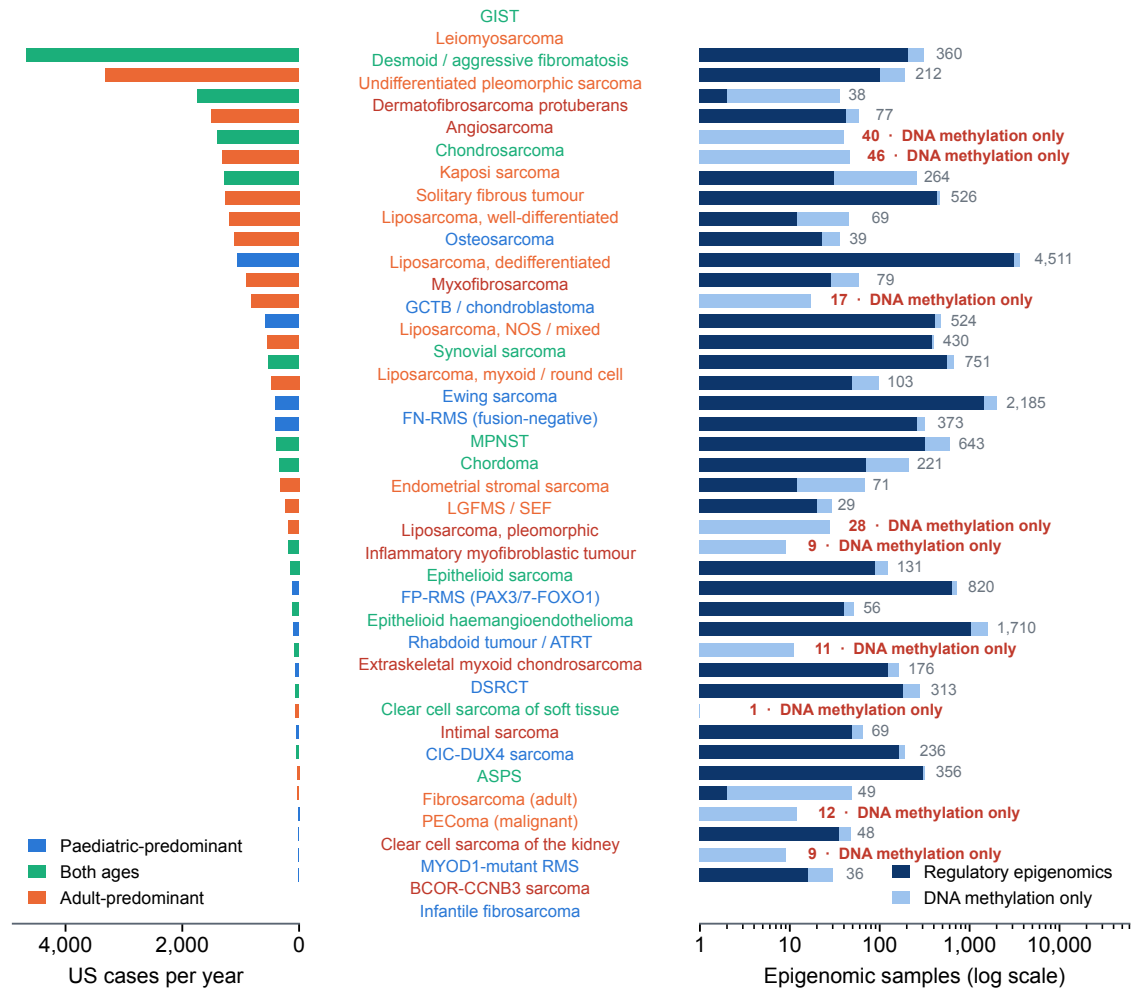
Regulatory epigenomic samples in public archives
(ChIP-seq / CUT&RUN / ATAC / DNase / Hi-C / HiChIP)

No regulatory epigenomics whatsoever

- Dermatofibrosarcoma protuberans (1,400/yr)
- Angiosarcoma (1,312/yr)
- Myxofibrosarcoma (820/yr)
- Liposarcoma, pleomorphic (185/yr)
- Inflammatory myofibroblastic tumour (175/yr)
- Extraskeletal myxoid chondrosarcoma (75/yr)
- Intimal sarcoma (60/yr)
- Clear cell sarcoma of the kidney (20/yr)
- BCOR-CCNB3 sarcoma (15/yr)

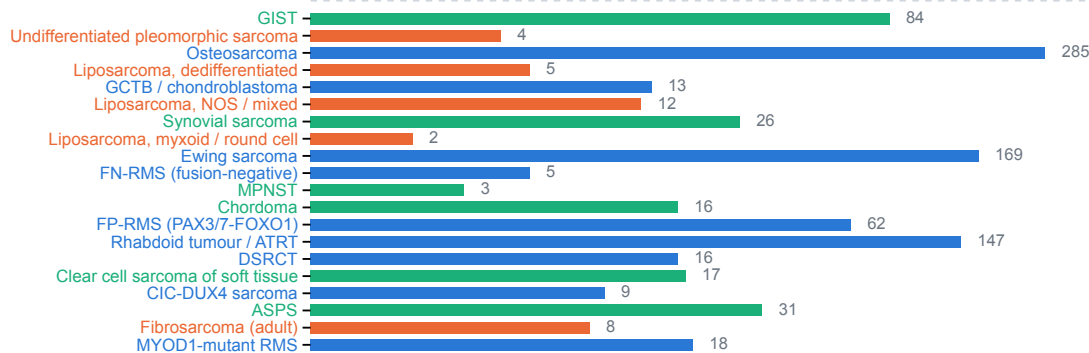


Nine sarcoma entities with a published incidence rate — two of them above 1,300 US cases a year — have DNA methylation profiling and nothing else Data, split by what it can answer



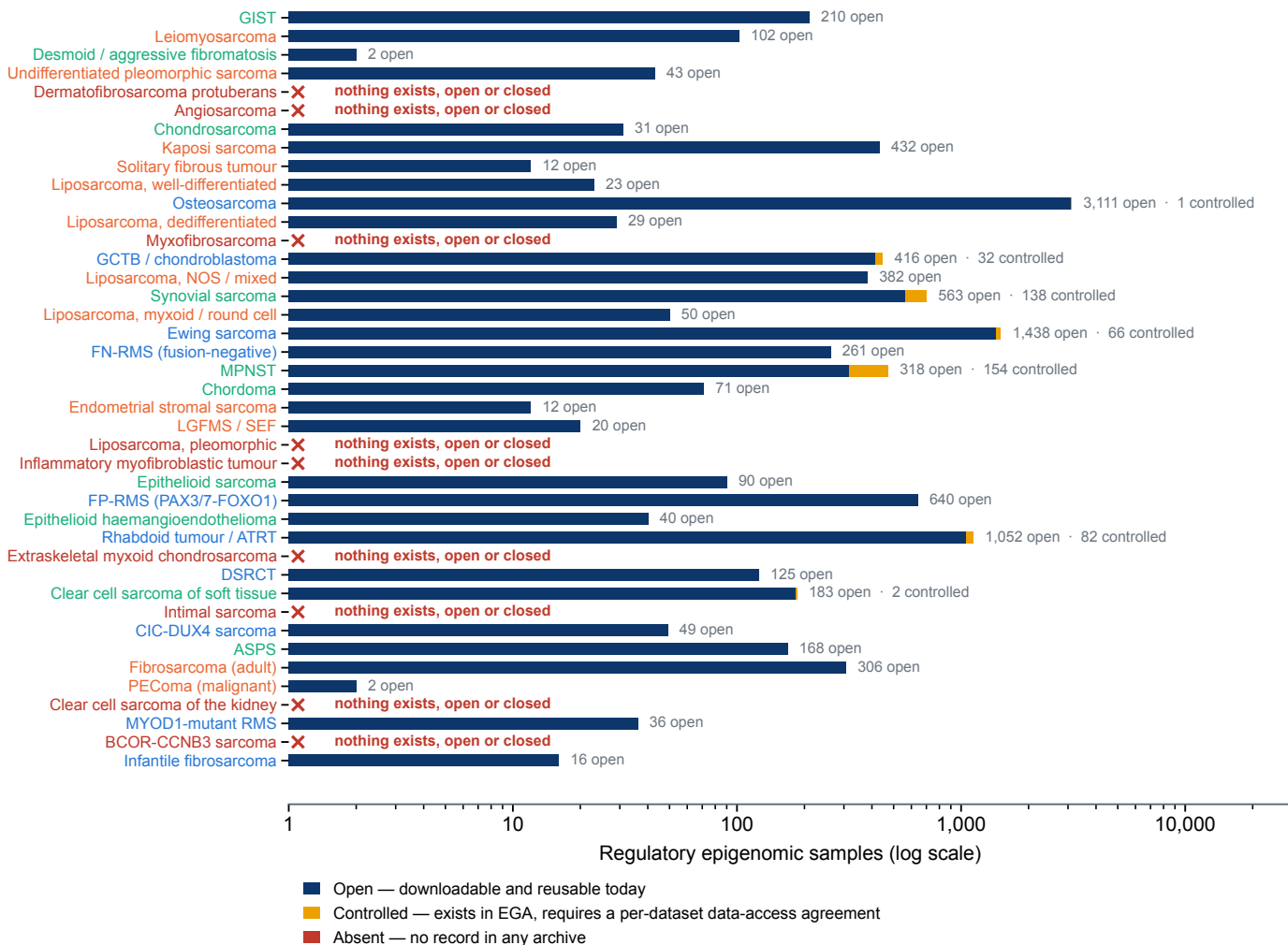
25 of 45 sarcoma entities have never had a single patient's tumour profiled for active chromatin, chromatin accessibility or 3D genome architecture

Leiomyosarcoma	✗	none, against ~3,310 US cases per year
Desmoid / aggressive fibromatosis	✗	none, against ~1,740 US cases per year
Dermatofibrosarcoma protuberans	✗	none, against ~1,400 US cases per year
Angiosarcoma	✗	none, against ~1,312 US cases per year
Chondrosarcoma	✗	none, against ~1,280 US cases per year
Kaposi sarcoma	✗	none, against ~1,268 US cases per year
Solitary fibrous tumour	✗	none, against ~1,200 US cases per year
Liposarcoma, well-differentiated	✗	none, against ~1,100 US cases per year
Myxofibrosarcoma	✗	none, against ~820 US cases per year
Endometrial stromal sarcoma	✗	none, against ~328 US cases per year
LGFMS / SEF	✗	none, against ~229 US cases per year
Liposarcoma, pleomorphic	✗	none, against ~185 US cases per year
Inflammatory myofibroblastic tumour	✗	none, against ~175 US cases per year
Epithelioid sarcoma	✗	none, against ~156 US cases per year
Epithelioid haemangi endothelioma	✗	none, against ~110 US cases per year
Extraskeletal myxoid chondrosarcoma	✗	none, against ~75 US cases per year
Intimal sarcoma	✗	none, against ~60 US cases per year
PEComa (malignant)	✗	none, against ~25 US cases per year
Clear cell sarcoma of the kidney	✗	none, against ~20 US cases per year
BCOR-CCNB3 sarcoma	✗	none, against ~15 US cases per year
Infantile fibrosarcoma	✗	none, against ~13 US cases per year
Myoepithelial carcinoma of soft tissue	✗	none (no population rate published anywhere)
Ossifying fibromyxoid tumour	✗	none (no population rate published anywhere)
Angiomatoid fibrous histiocytoma	✗	none (no population rate published anywhere)
Atypical fibroxanthoma / pleomorphic dermal sarcoma	✗	none (no population rate published anywhere)



Regulatory epigenomic profiles of PATIENT-DERIVED MATERIAL — tumour tissue, PDX or patient-derived organoids (ChIP-seq / CUT&RUN / ATAC / Hi-C; input and IgG controls excluded)

Some of the gap is unmeasured biology; some of it is data that exists and cannot be reused 1,207 regulatory epigenomes sit behind data-access committees — but not one covers the nine entities with nothing at all



A further 732 regulatory epigenomes sit in EGA under controlled access in cohorts whose subtype EGA does not state — RMS (any subtype): 581; Sarcoma, unspecified / mixed cohort: 151. The largest is the St Jude CSTN rhabdomyosarcoma ChIP-seq (EGAD00001004312 + EGAD00001006398), none of which is in GEO. They are excluded from the bars rather than attributed by guesswork.

Most sarcoma entities rest on one deposit — and for many of them that deposit carries no entity label in GEO at all

